Women’s Labour Market Diagnostic in India

From labour-market data to evidence for decision-making | Applied data analytics

Data source: Periodic Labour Force Survey (PLFS), accessed through NITI Aayog’s National Data and Analytics Platform (NDAP).

Problem and research question

Women’s labour-market outcomes vary across Indian states, education levels, rural–urban settings and employment structures. A single national participation rate cannot show these differences. This project asks: **How do women’s labour-market outcomes vary across these contexts, and what does the employment structure look like within them?**

Data and analytical approach

The project analyses two PLFS datasets: labour-market outcomes by education and employment structure by enterprise type and industry. The analysis covers 2017–2023 and combines data-quality checks, reusable analytical datasets, exploratory analysis, state and gender comparisons, statistical tests and effect sizes, 36-fold leave-one-state-out machine-learning validation, and guarded one-year forecasting. Findings are presented in a six-tab interactive dashboard.

Key findings

- **Gender gap persists:** The unweighted mean female Labour Force Participation Rate (LFPR) across 36 States/UTs rose from 25.78% in 2017 to 45.01% in 2023. Female LFPR remained below male LFPR in all 251 matched State–Year observations; the 2023 gap was about 33.1 percentage points.
- **Rural–urban divergence widened:** Mean rural female LFPR increased by about 24.5 percentage points between 2017 and 2023, compared with 9.4 points in urban areas.
- **Education and unemployment:** Participation varied non-linearly by education. In 2023, female graduate unemployment was 23.31%, compared with 10.74% among male graduates.
- **Predictive limits:** Under 36-fold leave-one-state-out cross-validation, Gradient Boosting achieved $R^2 = 0.155$, MAE = 15.70 percentage points and RMSE = 22.27 percentage points, indicating limited generalisation to an unseen state.
- **Exploratory forecast:** The eligible Holt forecast estimated female LFPR at 46.36% for 2024, with a nominal 80% interval of 43.73%–48.99%. Historical MAE was 2.21 percentage points versus 3.37 for the naïve baseline.

Relevance to NGOs and CSR organisations

The diagnostic can help organisations strengthen proposal problem statements, explore which geographies or groups warrant closer assessment, frame programme-design questions, and select relevant outcome indicators. It can inform consideration of interventions such as employability training, career guidance or employment support, subject to local needs assessment. State-level patterns alone cannot determine funding priorities or prove which intervention will work; local data, community consultations and employer perspectives remain essential.

Limitations and responsible interpretation

LFPR figures are unweighted means across States/UTs, not official population-weighted national estimates. The analysis is descriptive and observational, does not establish causation or programme impact, and the enterprise analysis has coverage limitations. Forecasts are exploratory, based on only seven annual observations, and subject to substantial uncertainty.

Project links

Static dashboard | Gradio dashboard | GitHub repository (verify URL before submission)